

Chapter 11

Antialiasing

Originally prepared by Dr Kristian Spoerer of UNUK

Modified by Prof Sean He of UNNC

Learning Outcomes

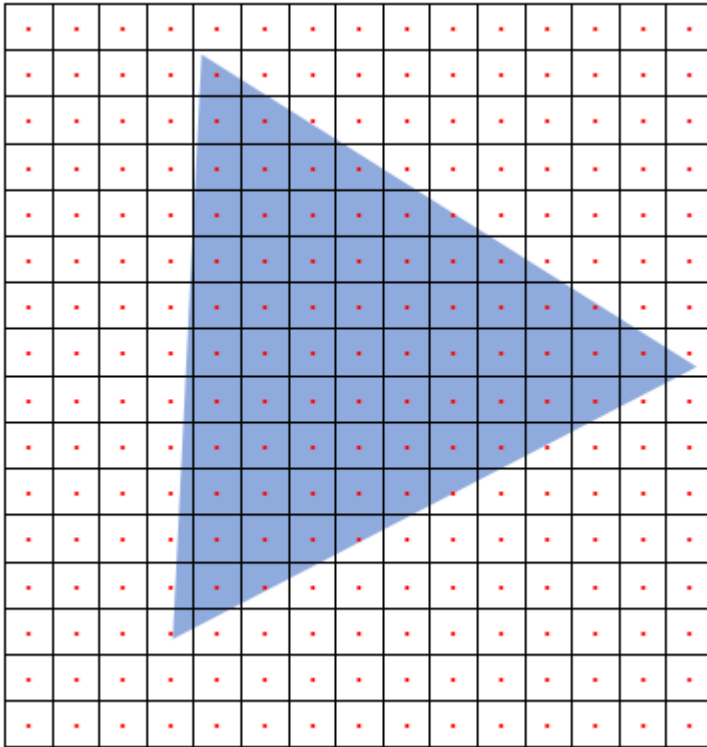
After studying this chapter, you are expected to be able to

1. Define aliasing and antialiasing.
2. Recognise the manifestation of aliasing in the form of 'jaggies'.
3. Explain what causes 'jaggies'.
4. Describe three antialiasing algorithms, Supersampling, Multisampling, and Temporal antialiasing.
5. Describe the similarities and differences between the way the three antialiasing algorithms work.
6. Recognise the smoothing effect of antialiasing which removes 'jaggies'.
7. Recognise that there is no way to absolutely solve aliasing by increasing samples.
8. Explain the positives and negatives of the three antialiasing algorithms.
9. Explain the difference between 2x and 4x antialiasing.
10. Compare the time requirements of the antialiasing algorithms.
11. Critically evaluate the effectiveness of the antialiasing algorithms at removing the 'jaggies' caused by aliasing.

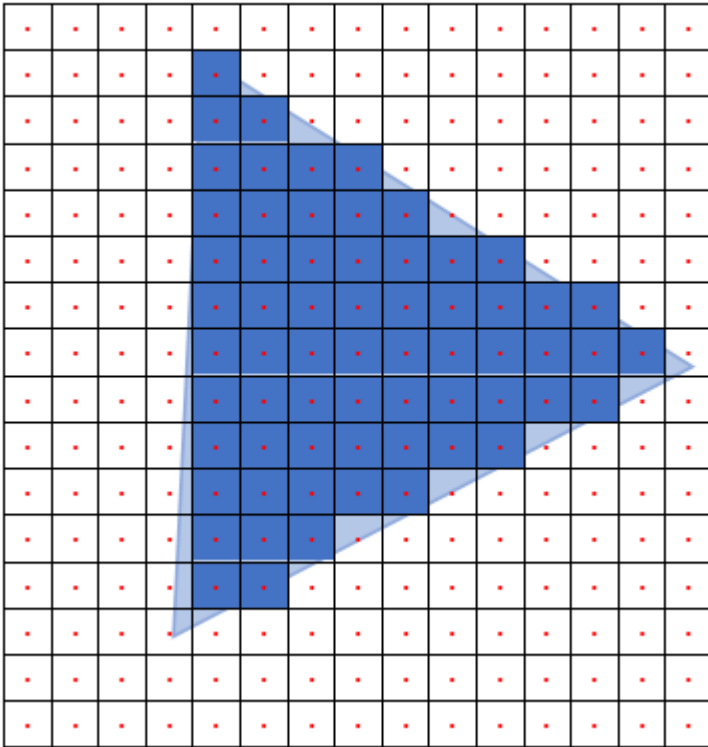
Aliasing

Rendering essentially boils down to sampling. A display is an array of *width* x *height* pixels, and each pixel needs to be sampled to decide the colour that it should be displayed. This sampling involves calculations in a 3-dimensional space.

Aliasing is an undesirable effect of the frequency chosen to take samples. More specifically, aliasing happens when samples are taken at a frequency which is too low.



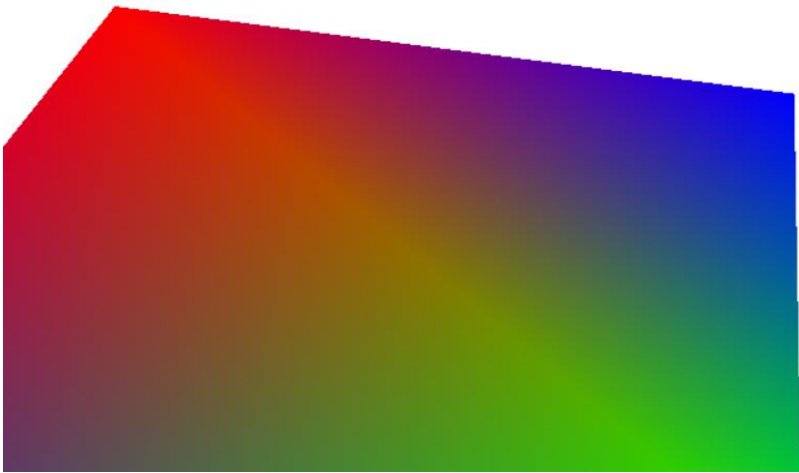
The above image shows an array of 15 x 16 pixels with a red sample point at the centre of each pixel and a blue triangle which is going to be rendered.



If the samples are taken only at the pixel centre points then the resulting colours will be as shown as in the figure above. Note that the colours of each pixel are darker than the colour of the triangle just for illustration purpose and to be able to see both triangle and pixels together. The actual pixels would be coloured in as the colour returned by the sampling procedure. Clearly, information is lost in the sampling process.



As can be seen in the figure above, there are parts of the triangle which are not sampled. The triangle will be rendered with clear aliasing problems.



This aliasing is often called “jaggies”, and they can be seen along the edges of the polygons in the figure above.

The remainder of this chapter will cover methods for reducing the effects of aliasing, which are called antialiasing techniques.

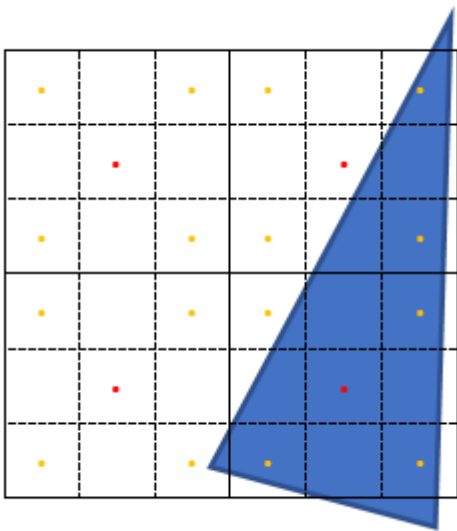
Supersampling

Supersampling, also called *Supersampling Antialiasing SSAA*, is the most obvious idea for how to reduce aliasing.

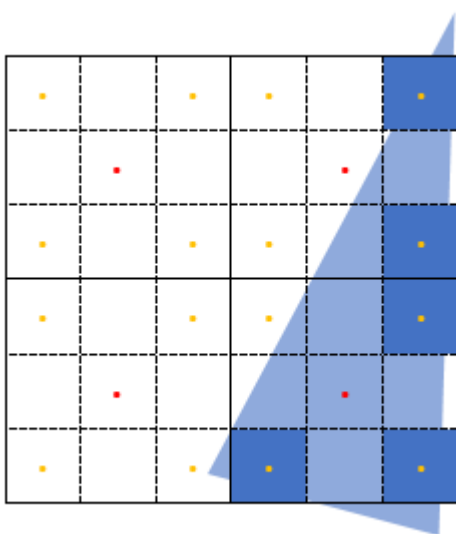
If aliasing happens because there are too few samples, then it is a natural idea to perform more samples.

Supersampling can be modified by the number of samples taken in each dimension for one pixel. 2x Supersampling generates twice as many fragments in the x and y screen dimensions. 4x Supersampling generates four times as many fragments in the x and y screen dimensions.

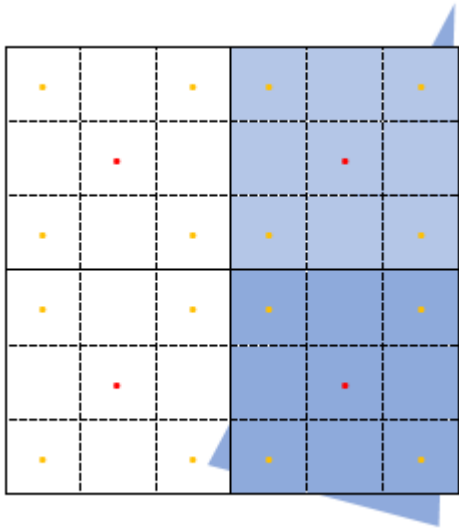
If the display resolution is 600 by 400 pixels, 2x Supersampling works by generating twice as many fragments in the x and y screen dimensions, sampling the fragment colour using the fragment shader, and then storing the colours in a colour buffer which is 1200 x 800 in size. Then, when the pixels are displayed, the colour of each pixel is the average colour of the 2 x 2 samples that make up the colour buffer.



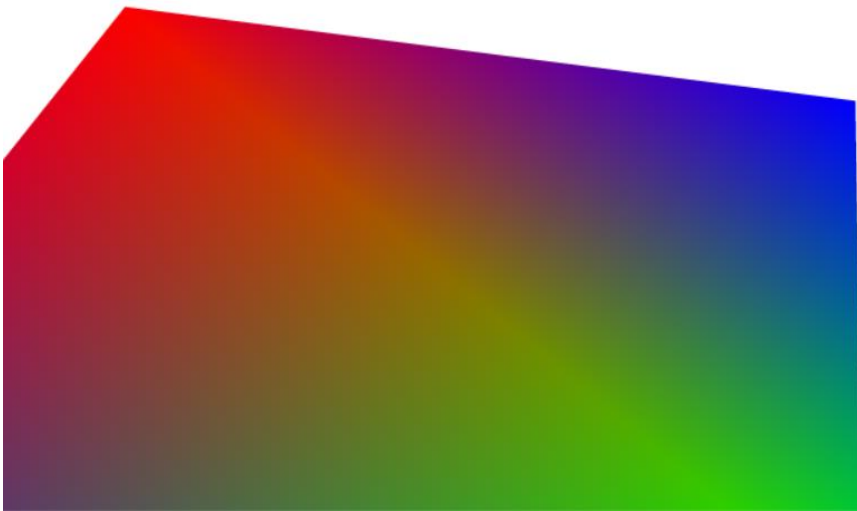
In this figure, 4 pixels can be seen with the centre points in red.



Each pixel has 4 samples taken using the fragment shader as shown by the points in yellow.



Each pixel colour is then calculated as the average of the 4 samples for that pixel.



The resulting render which has been Supersampled is much smoother and the jaggies have been remedied, as can be seen in the image above. This image was rendered using 4x Supersampling.

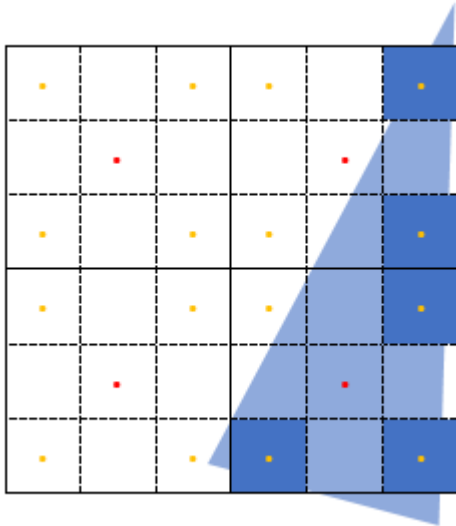
Supersampling was provided as a hardware solution for the Sega Dreamcast.

However, Supersampling is extremely expensive to compute because all samples are coloured by the return value of the fragment shader. In 4x Supersampling, each pixel has 16 fragments shaded.

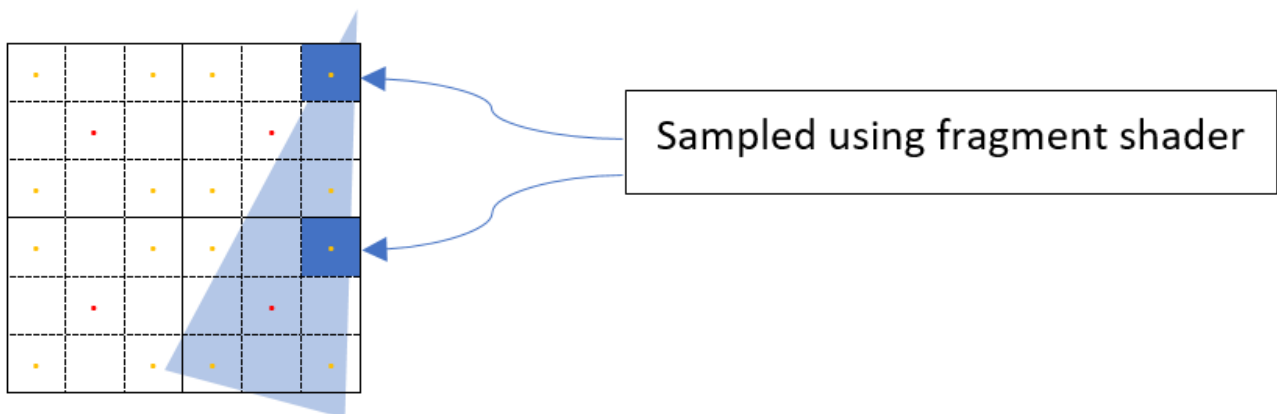
Multisampling

Multisampling, known as *Multisampling Antialiasing MSAA*, is an attempt at increasing the number of samples but without calling the fragment shader as many times as is called in Supersampling.

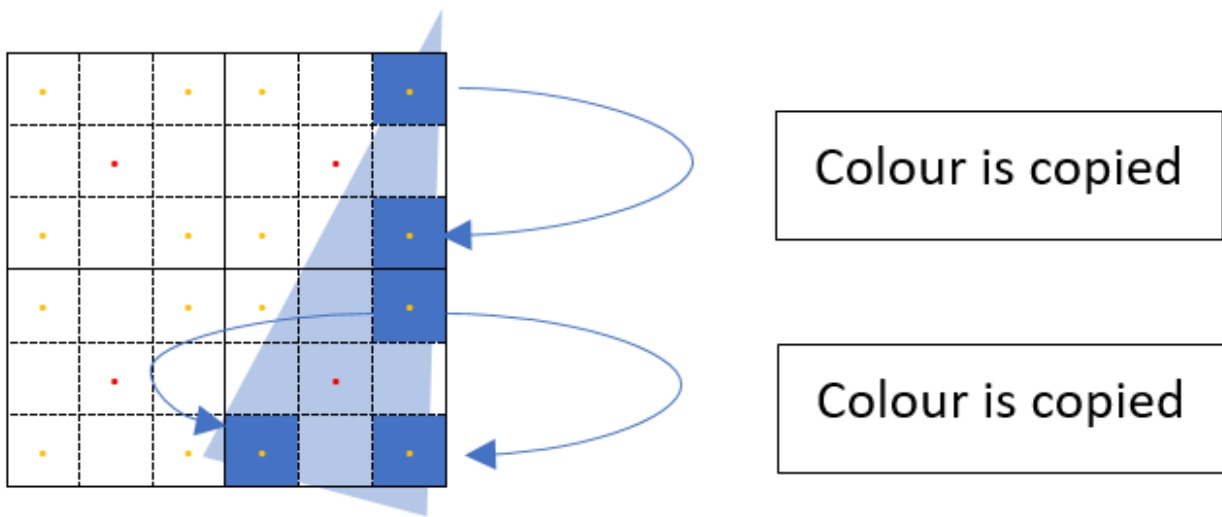
In Supersampling, the fragment shader is called one time for **each sample** that is inside the triangle



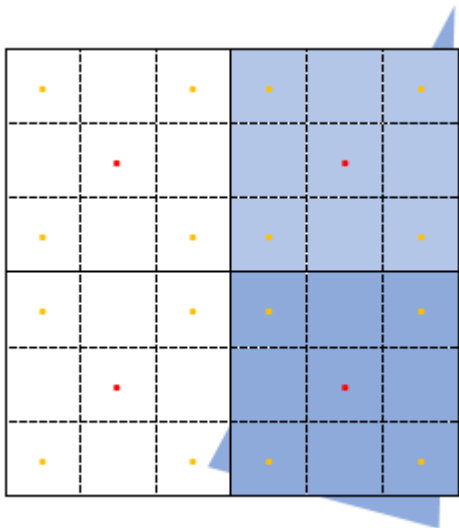
In Multisampling, the fragment shader is only called one time for **each pixel** that is inside the triangle.



and the colour is copied to the sample points that are inside the triangle.



The pixel colours are then the average of the 4 samples.

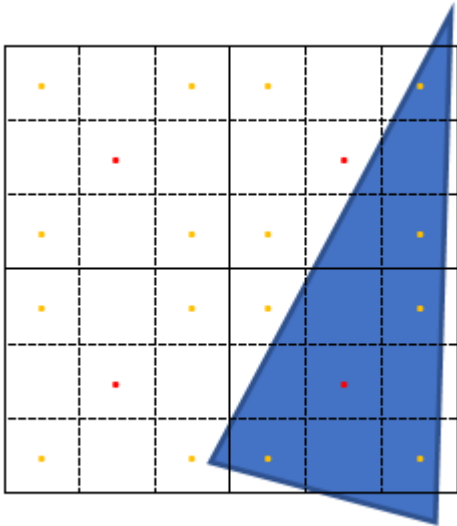


The saving with MSAA compared to SSAA is that the fragment shader is called $2 \times 2 = 4$ times for each pixel for each triangle in SSAA, whereas the fragment shader is only called once for each pixel for each triangle for MSAA.

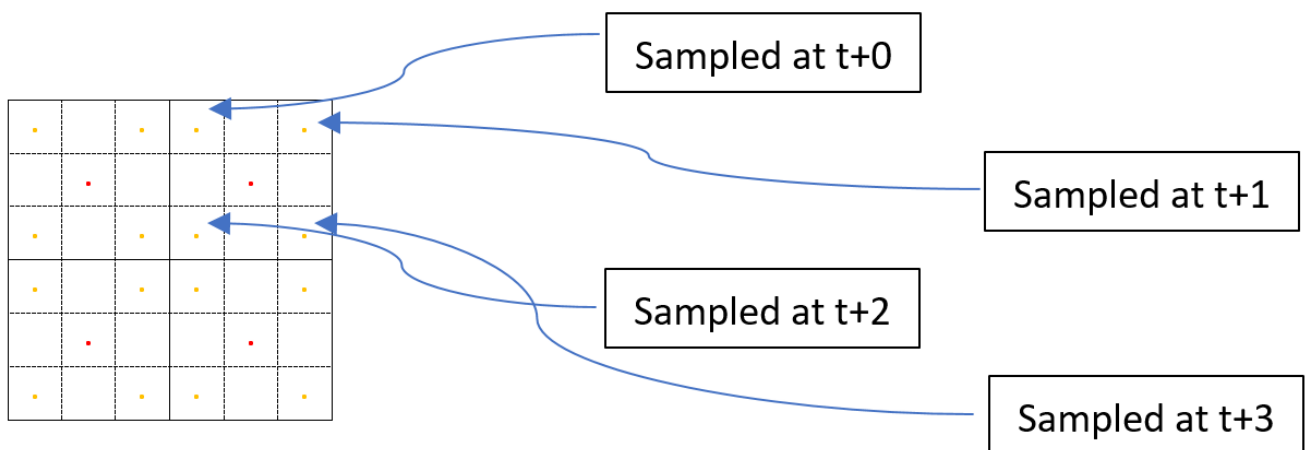
Temporal Antialiasing

Temporal Antialiasing TAA is another antialiasing algorithm. Temporal Antialiasing works in much the same way as Supersampling.

If the display, resolution is 600 by 400 pixels with 2x Temporal Antialiasing, the colours are stored in a colour buffer which is 1200 x 800 in size. Then, when the pixels are displayed, the colour of each pixel is the average colour of the 2 x 2 samples that make up the colour buffer. This is the same as 2x Supersampling.



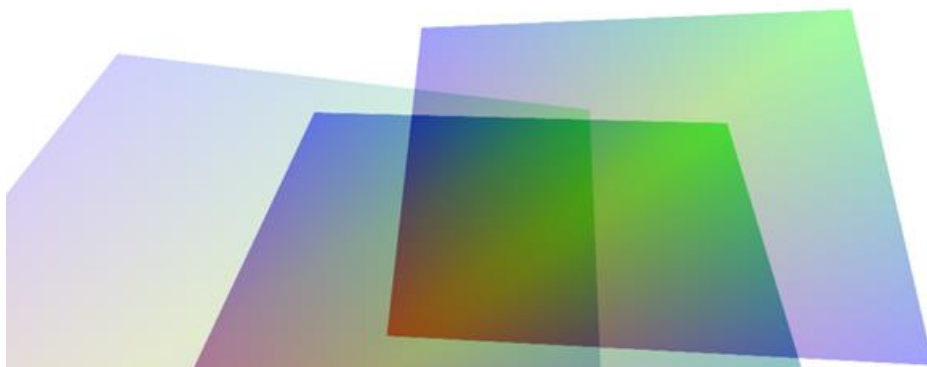
The difference is that, with TAA, the 4 samples for each pixel are not generated by the fragment shader every frame. Instead, the TAA algorithm takes one sample per frame, and therefore requires four frames (or time points) to take the four samples.



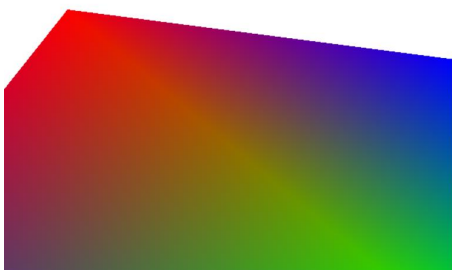
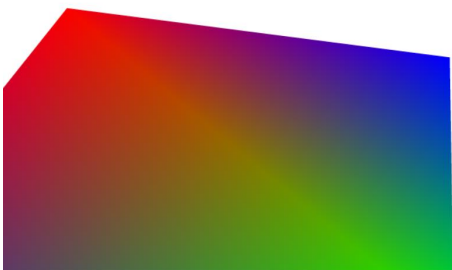
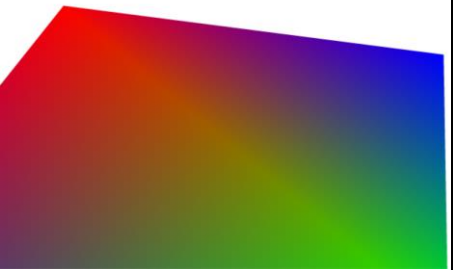
This means that the number of samples in the x and y screen dimensions is 1 per pixel per frame, which is the same as if no anti-aliasing is used. When the pixels are displayed the colour of each pixel is the average colour of the 2 x 2 samples that make up the colour buffer. Which means that the effect is the same as if 2x Supersampling is used.

This means there is the antialiasing effect of Supersampling but without the large time requirement.

The downside of Temporal Antialiasing is that there are ghost artifacts if the image moves fast.



If a static image is rendered then the effect of Temporal Antialiasing is the same as Supersampling, but TAA is much faster.

		
No Antialiasing	Supersampling	Temporal Antialiasing
x 1	x 9	x 1.95

In this figure, the jaggies caused by aliasing can be seen in the left image. In the middle image, Supersampling has removed the aliasing but has taken approximately 9 times longer to render the frame. In the right image, Temporal Antialiasing produces the same visual effect as Supersampling, but has only taken 1.95 times longer to render the frame compared to no Antialiasing.

The additional time required by Temporal Antialiasing is because each triangle needs to be translated by fraction of a pixel before the sample is taken and then translated back by the same fraction of a pixel after the sample is taken.